

Rupjyoti Barman

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Summary

.NET Backend Developer with **3 years** of experience building scalable backend applications, system-level diagnostic platforms, and web solutions for **HP**. Skilled in **C#, ASP.NET Core, Web API, Entity Framework Core, and React**, with experience designing decoupled architectures, integrating native C++ components, and developing enterprise-grade solutions. Proven ability to deliver high-impact features, optimize system performance, and collaborate with global engineering teams across India, Taiwan, and the US.

Skills

- **Languages:** C#, C++, SQL, JavaScript, Python
- **Frameworks & Libraries:** ASP.NET Core (Web API, MVC), Entity Framework Core, WPF, React, Bootstrap, jQuery, LINQ
- **Windows APIs & Interop:** WMI, WQL, ManagementEventWatcher, WUA API, HPCMSL, DeviceIoControl, P/Invoke, Win32 APIs
- **Architecture & Design:** Clean Architecture, Layered Architecture, SOLID Principles, OOP, Factory Pattern, Dependency Injection, Controller-Service-Repository Pattern
- **Database, DevOps & Tools:** SQL Server, PostgreSQL, EF Core (Code-First, Migrations), CI/CD, Azure Pipelines, Git, NuGet, Visual Studio, Postman, Jira, Confluence, Serilog
- **Testing & Security:** MSTest, Moq, JWT Authentication, RBAC

Experience

Software Developer, BoldTek India Pvt. Ltd. - Guwahati

June 2023 - Present

Client: Hewlett-Packard (HP)

- Engineered 2 production features in an **ASP.NET Core MVC** application (Controller-Service-Repository, **EF Core Code-First**, SQL Server, JavaScript, Bootstrap); built an **ODM self-service dashboard** with 3-tier RBAC (Viewer, ODM, Admin) for INI configuration management, verification submission, and approval tracking with INI key-value validation across **1000+ HP NPI config files**; developed a **Release Notes Dashboard** replacing manual Excel-based tracking for the HP Factory Diagnostic team.
- Built a **React-based Diagnostic Release Tracking Portal** (POC) using **React Hooks, React Router**, and reusable components with an **ASP.NET Core Web API** backend; centralized tracking of diagnostic releases, NuGet package versions, issue resolutions, and development milestones.
- Led the India team in migrating from a **shared RPC-based Windows Service architecture** to a **decoupled three-layer NuGet-based architecture** (Business Logic, C# Interop Wrapper via P/Invoke, Native C++ DLL) with **Factory Pattern**-based dependency injection; achieved **100% elimination of third-party release delays** (previously 10–15 days) and **reduced, diagnostic development cycle** from 4 to 3 sprints (40 to 30 working days, **25% reduction**) by removing 2 redundant architectural layers, resolving cross-team unhandled exceptions and sleep and hibernate-related service failures in **HP PC Hardware Diagnostics**.
- Migrated **6 hardware diagnostic modules** (**Storage:** DRV Read/Verify, SMART Health Monitor, SMART Data Collection; **Network:** NIC IRQ Verification, ROM/MAC Validation; **Audio:** Audio Playback Test) onto the new Decoupled Architecture; additionally **engineered 2 diagnostic modules** from scratch - **Driver Update Discovery** (WUA API) and **BIOS Update Check** (HPCMSL, commercial systems) - consolidating all into a **centralized diagnostic backend** supporting multiple HP client applications with zero external service dependency.
- Replaced a service-based device discovery architecture (requiring a 2 to 3 minute service restart per hardware change) by **engineering a real-time WMI monitoring module** (ManagementEventWatcher and WQL on Win32_PnPEntity) into the shared "**Common**" library; reduced detection latency **from 2 to 3 minutes to under 4 seconds (97% reduction)**, adopted across multiple hardware diagnostic APIs.
- Configured **Azure Pipelines** to trigger **NuGet package builds**, publishing the Business Logic package (with dependencies) into the .NET Wrapper layer to produce the final consolidated NuGet package (bundling all layers including Native C++ DLL) for testing and client consumption.
- Verified diagnostic modules using a **WPF-based API testing application** for end-to-end UI-level verification alongside **MSTest** unit test projects across **Business Logic** and **.NET Wrapper** layers, leveraging **Moq** for dependency mocking to ensure reliability prior to client integration.
- Partnered with cross-functional global teams (**India, Taiwan, US**) to deliver features, debug production issues, and align on diagnostic API design across release cycles.

Education

B.Tech in Computer Science and Engineering - CGPA: 8.01/10

May 2023

North Eastern Regional Institute of Science and Technology (NERIST), Arunachal Pradesh